API Guide

The API to the energidataservice.dk platform enables users free of charge to download data for further processing.

Best practices

The platform does not have unlimited resources and do have limitations as described below.

Not a direct source for applications

It is not meant as the source for all units for vendor applications. In this situation the vendor must download data and then be the source of all possible units for the application.

Limits on Requests and Update Frequencies

The number of requests should correspond to the update frequency of the datasets.

General guideline:
1 request per Update Frequency using a dynamic timestamp start=now-(3x Update Frequency).
e.g.:
For datasets updated every 5 minutes: 1 request every 5 minutes with parameters start=now-PT15M

```
https://api.energidataservice.dk/dataset/CO2Emis?start=now-PT15M
```

Update frequency details are listed on the dataset pages.

For some datasets special recommendations are noted on the respective dataset pages.

A maximum of 1 request per unique IP address per dataset per minute is allowed.
IP addresses will be blocked if an excessive number of requests is observed. In the future restrictions will be technically implemented.

Dynamic timestamps

Request URLs can be optimized using dynamic timestamps. Details are explained in the section Parameters.

Build your URL

Request methods

The restful API is a synchronous interface with Http Get. Parameters are a part of the URI string.
To get data:

```
https://api.energidataservice.dk/dataset/{name of dataset}?
```

or to get metadata:

```
https://api.energidataservice.dk/meta/dataset/{dataset | group | organization}?
```

followed by Parameter Name = Parameter Value, where each pair of parameter name and value are separated by &.

Build your request URL

To get a graphical guide to the API, you can access our swagger documentation.

Query the API

The Energi Data Service platform web API can be accessed via web browser using the Get method. The following example returns data for Price area = DK1 for May, 2022 for the DeclarationProduction dataset:

```
https://api.energidataservice.dk/dataset/DeclarationProduction?start=2022-05-01&end=2022-06-01&filter={"PriceArea":["DK1"]}
```

Below the description of Parameters and Limits you can see more examples of requests and their output using the different parameters.

Parameters

Only equal operator is available for all parameters.
Both ASCII characters and URL-encoding forms may be used.

`start`
Specifies the start point of the period for the data request. The period is selected based on the column given by the Time range column, which is specific to the dataset. The start point is included in the output. Example: "start=2022-07-01".

Datetime formats which may be used: "yyyy", "yyyy-MM", "yyyy-MM-dd", "yyyy-MM-ddTHH:mm"

The date and time refer to the Danish timezone / local Danish time.

Dynamic timestamps
You can also provide a dynamic timestamp that is relative to the current point in time. To do so the value should follow this format: `{now | StartOfDay | StartOfMonth | StartOfYear}{+(written as %2B)) | -}{ISO_8601 duration}`. "start=now" will set the starting point to now. "start=now-P1D" will set the starting point to 1 day before now Danish time. For timestamps in the future, write %2B instead of +. For instance start=now%2BP1D will set the starting point to 1 day after now Danish time.

`end`
Specifies the end point of the period for the data request. The period is selected based on the column given by the Time range column. The end point is excluded from the output. Example: "end=2022-08-01".

`columns`
Optional - comma separated list of columns. If left blank all columns are included. Example: "columns=HourUTC,PriceArea,CO2Emis".

`filter`
Json object of key/value pairs where keys are column names and values are json arrays of possible values for that column.
Example: `filter={"DK36Code":["QB"],"DK19Code":["Q"]}` will return data where both conditions are met.
Example: `filter={"PriceArea":["DK1","DK2"]}` will return data where "PriceArea" is either "DK1" or "DK2".

Single quotes in strings: In order to filter values containing single quotes, replace the single quote (') with two quotes ('').

`sort`
Comma separated list of columns to sort by. Example: "sort=HourUTC desc,PriceArea". If left blank the output is sorted descending according to the unique key (e.g. HourUTC or Minutes5UTC) by default. If the sort parameter is set, the default sort order is ASC (ascending). E.g. "sort=CO2Emission" will sort ascending and "sort=CO2Emission desc" will sort descending.

`offset`
Number of records to skip. Example: "offset=30".

`limit`
Maximum number of records to return. Example: "limit=30".
"limit=0" returns all records. If neither "start" nor "limit" is provided a default value of 100 is used.

Examples

Below you can see examples of requests and their output using the different parameters.
Note that limit is added in all requests to limit the outputs of the examples. To use the request URL without limitations remove "&limit=4".
Also note that the outputs are displayed in descending order according to the unique key (e.g. HourUTC or Minutes5UTC) by default.

`start and end`
To get the first hour of January 1st 2022 from the dataset CO2 Emission:

```
https://api.energidataservice.dk/dataset/CO2Emis?start=2022-01-01T00:00&end=2022-01-01T01:00&limit=4
```

View output example

`Start of year as start point`
Using the startOfYear value as a starting point:

```
https://api.energidataservice.dk/dataset/CO2Emis?start=StartOfYear&end=now&limit=4
```

View output example

`Dynamic start and end`
To get the last year from the dataset CO2 Emission:

```
https://api.energidataservice.dk/dataset/CO2Emis?start=now-P1Y&end=now&limit=4
```

View output example

`columns`
To only get the columns Minutes5DK, PriceArea and CO2Emission from the dataset CO2 Emission:

```
https://api.energidataservice.dk/dataset/CO2Emis?columns=Minutes5DK,PriceArea,CO2Emission&limit=4
```

View output example

`filter`
To get data from either DK1 or DK2 from the dataset CO2 Emission:

```
https://api.energidataservice.dk/dataset/CO2Emis?filter={"PriceArea":["DK1","DK2"]}&limit=4
```

View output example

`sort`
If set default sort order is ascending.

To sort by CO2 emission value in descending order:

```
https://api.energidataservice.dk/dataset/CO2Emis?sort=CO2Emission desc&limit=4
```

or (the space between the value and sort order may be replaced with %20 as Both ASCII characters and URL-encoding forms may be used):

```
https://api.energidataservice.dk/dataset/CO2Emis?sort=CO2Emission%20desc&limit=4
```

View output example

`sort by two columns`
To sort by Price area and CO2 emission:

```
https://api.energidataservice.dk/dataset/CO2Emis?sort=PriceArea,CO2Emission&limit=4
```

View output example

`Combination of parameters start, end, columns, filter, sort and limit`
To get only the first 4 rows of data from the first hour of January 1st 2022, only the columns Minutes5DK, PriceArea and CO2Emission and only PriceArea DK1, sorted by CO2Emission value in descending order:

```
https://api.energidataservice.dk/dataset/CO2Emis?start=2022-01-01T00:00&end=2022-01-01T01:00&columns=Minutes5DK,PriceArea,CO2Emission&filter={"PriceArea":["DK1"]}&sort=CO2Emission desc&limit=4
```

View output example

`offset`
To not include the first 10 rows of the dataset CO2 Emission:

```
https://api.energidataservice.dk/dataset/CO2Emis?offset=10&limit=4
```

`timezone`
To get the first hour of January 1st 2022 from the dataset CO2 Emission using UTC time zone:

```
https://api.energidataservice.dk/dataset/CO2Emis?start=2022-01-01T00:00&end=2022-01-01T01:00&timezone=UTC&limit=4
```

View output example

Download URL

It is possible to download data directly in the formats xlxs, json or csv by adding download?format=filetype after the name of the dataset and before the chosen parameters.

To download Excel: download?format=XL

```
https://api.energidataservice.dk/dataset/CO2Emis/download?format=XL&limit=10
```

To download json: download?format=json or download? (as json is default)

```
https://api.energidataservice.dk/dataset/CO2Emis/download?format=json&limit=10
```

To download csv: download?format=csv

```
https://api.energidataservice.dk/dataset/CO2Emis/download?format=csv&limit=10
```